Dalhousie University

CSCI 4144 – Data Mining and Data Warehousing

Project Title:

Implementing and Evaluating a Sliding Window Classifier with Drift Detection for Learning under Concept Drift

Group ID: [1]

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Introduction:

When the underlying distribution of data shifts over time, previously trained machine learning models become less accurate. The occurrence is known as concept drift. Data is frequently not stationary in real-world applications, which means that external factors like user behavior, environmental changes, or adversarial acts cause the connection between input attributes and target labels to vary. As per [1], concept drift happens when the source of the data changes over time, resulting in a gap between training data from the past and observations from the future.

Traditional machine learning models assume that training and testing data follow the same distribution. However, this assumption does not hold in many real-world scenarios, resulting in reduced performance as time progresses. To address this issue, concept drift learning frameworks adopt an incremental learning approach, where models are continuously updated using newly observed data. At each time step, the model is trained on historical data and used to predict the next instance, after which the true label becomes available and is incorporated into future training [1].

In this project, we use two essential elements - a sliding window and a drift detection mechanism (DDM) - to create an adaptive learning system intended to manage concept drift. By keeping some of the most recent data, the sliding window makes sure that out-of-date information is progressively removed. In the meantime, the drift detection technique keeps an eye on prediction mistakes and alerts users when a significant change has taken place. When combined, these elements enable the model to dynamically adjust to shifting data distributions, enhancing performance in non-stationary settings.

Algorithm:

This project implements two core components to handle concept drift in a data stream: a Sliding Window classifier and the Drift Detection Method (DDM).

Sliding Window

A sliding window maintains a fixed-size buffer of the most recent training instances. As new labeled instances arrive in the stream, they are added to the window. When the window exceeds its maximum size, the oldest instance is removed. This ensures that the classifier is always trained on the most recent data, gradually discarding outdated patterns that no longer reflect the current data distribution. In this project, a window size of 500 was used, with a minimum of 200 instances required before any predictions are made.

The sliding window addresses a fundamental challenge in concept drift, a model trained on all historical data weighs old and new patterns equally, making it slow to adapt. By restricting training to recent instances, the window ensures the classifier remains sensitive to distributional changes.

Drift Detection Method (DDM)

DDM, introduced by Gama et al. [2], is a trigger-based drift detection algorithm that monitors the classifier's online error rate as a binomial random variable. At each time step, the error is recorded as 1 if the prediction was incorrect and 0 if correct.

The running error rate p and its standard deviation s are updated incrementally as follows:

$$p = p + (e - p) / n$$

$$s = \sqrt{p(1 - p) / n}$$

where n is the total number of instances seen and e is the current prediction error. DDM tracks the minimum values of p and s observed so far, denoted p_min and s_min. These minimums represent the best performance achieved by the current concept. Two thresholds are defined relative to these minimums:

- **Warning level:** $p + s \geq p_min + 2 \times s_min$
- **Drift level:** $p + s \geq p_min + 3 \times s_min$

When the warning level is reached, DDM records the current stream index as the warning point and begins tracking data from that point onward. No immediate action is taken as the warning may represent noise rather than true drift. When the drift level is reached, drift is confirmed. The classifier's training window is reset to contain only the data collected since the warning point, ensuring the model retrains exclusively on instances that reflect the new concept. DDM's internal statistics are then fully reset to begin monitoring the next concept fresh.

This design makes DDM highly memory efficient, it never stores the full error history, only the running statistics and the current minimums. It is also self-calibrating, since the thresholds are relative to the best observed performance rather than fixed values, making it robust across different datasets and drift magnitudes.

Data Preparation and Algorithm Implementation:

Dataset

The ELEC2 dataset was used to evaluate the algorithm. It is sourced from the MOA streaming data repository and contains 45,312 instances across 6 input features and 1 class label. The features are: period (time of day normalized between 0 and 1), nswprice (electricity price in New South Wales), nswdemand (electricity demand in New South Wales), vicprice (electricity price in Victoria), vicdemand (electricity demand in Victoria), and transfer (scheduled electricity transfer between the two states). The class label is binary, 0 indicates the electricity price went down compared to the previous 24 hours and 1 indicates it went up. The dataset spans two years of real electricity market data, making it a well-established benchmark for concept drift evaluation due to the natural distributional shifts that occur over time as market conditions and demand patterns evolve.

Data Preparation

The dataset was loaded from a CSV file using pandas. Features and labels were separated into X and y respectively and converted to NumPy arrays for compatibility with scikit-learn. Crucially, the temporal order of instances was preserved throughout; no shuffling was applied. This is essential because concept

drift is a time-dependent phenomenon, and shuffling would destroy the distributional shifts that the algorithm is designed to detect.

Algorithm Implementation

data_loader.py: handles loading and preparing the ELEC2 dataset, separating features from labels and returning NumPy arrays in temporal order.

Ddm.py: implements the DDM algorithm, maintaining running error rate statistics and returning one of three states, stable, warning, or drift, at each time step.

sliding_window.py: maintains a fixed-size buffer of the most recent training instances, supporting instance addition, full reset on drift, and retrieval of data from a specific index onward for post-warning data recovery.

evaluator.py: tracks prequential accuracy and window accuracy over the stream, records warning and drift event indices, and generates plots with annotated drift and warning markers.

Main.py: serves as the entry point. It loads the data, initializes all components, and runs two experiments, one adaptive run using DDM and one static baseline run without drift detection, then prints a final comparison of results.

The classifier used is a Decision Tree from scikit-learn, retrained every 50 instances on the current window contents. A window size of 500 and a minimum training size of 200 instances were used for both experiments.

Experimental Results:

Two experiments were conducted on the ELEC2 dataset, an adaptive run using DDM-triggered window resets and a static baseline run with no drift detection. Both used a Decision Tree classifier with a window size of 500 and a minimum training size of 200 instances.

Results Summary

Metric	Adaptive (DDM)	Baseline
Total Instances Processed	16,456	45,112
Final Prequential Accuracy	80.39%	77.28%
Final Window Accuracy	77.90%	77.30%
Warnings Triggered	2,936	0
Drifts Detected	144	0

The adaptive classifier outperformed the static baseline by approximately 3 percentage points in final prequential accuracy, demonstrating that DDM-triggered resets meaningfully improve classification performance in the presence of concept drift.

Figure 1: Adaptive Classifier (DDM)

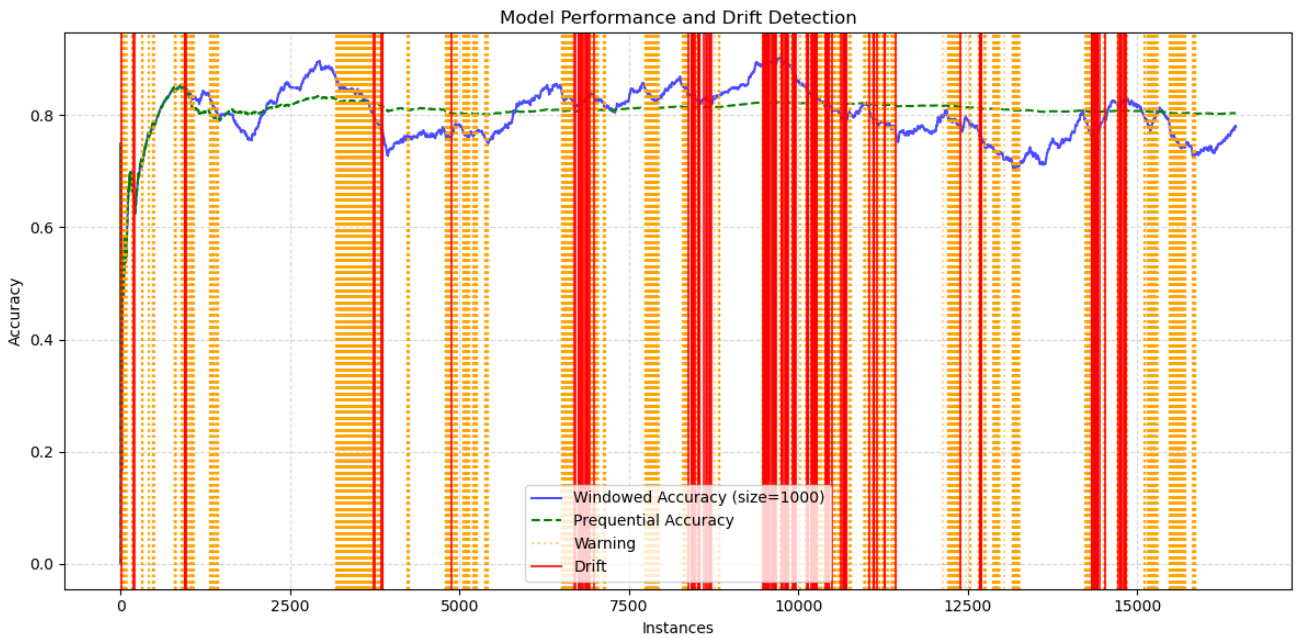

Figure 1 shows the adaptive classifier's performance over the stream. The windowed accuracy (blue) remains relatively stable around 80% throughout the stream despite the presence of concept drift. The high density of orange warning markers and red drift markers indicates that DDM is actively monitoring and responding to distributional changes. Each cluster of drift detections corresponds to a period of instability in the data, after which the window is reset and the classifier recovers.

Figure 2: Baseline Classifier (No DDM)

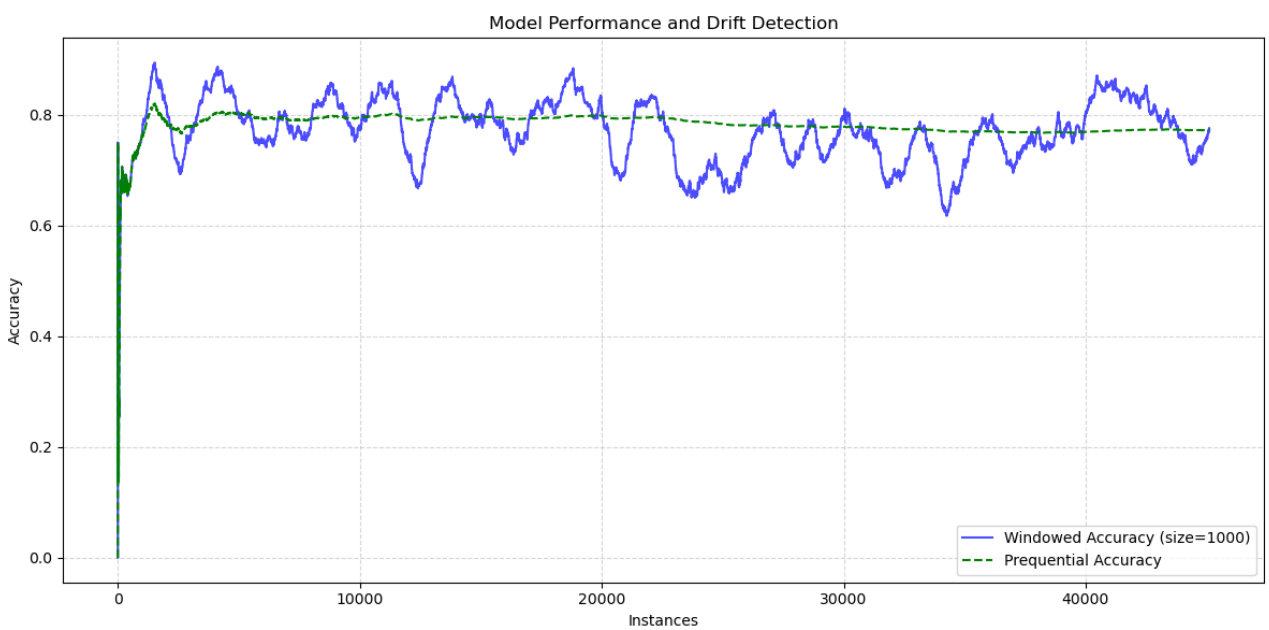

Figure 2 shows the baseline classifier's performance. With no drift detection mechanism, the classifier processes all 45,112 instances without any adaptation. The prequential accuracy (green dashed) shows a gradual declining trend over time, which is direct evidence of concept drift degrading model performance. The windowed accuracy (blue) fluctuates significantly, with notable drops around instances 12,000 and 35,000, corresponding to periods of distributional shift that the static model cannot adapt to.

Discussion

The results confirm that concept drift is present in the ELEC2 dataset and that DDM successfully detects and responds to it. The adaptive classifier maintains higher accuracy by continuously retraining on post-drift data, while the baseline degrades gradually as the data distribution shifts away from what the model originally learned.

One notable observation is that the adaptive run processed fewer total instances (16,456 vs 45,112). This occurs because frequent window resets after drift detection reduce the window size below the minimum training threshold of 200, causing some instances to be skipped until the window rebuilds. This is an inherent tradeoff of aggressive drift response, faster adaptation at the cost of some instances not being evaluated.

The high number of warnings (2,936) and drifts (144) suggests that DDM is sensitive to noise in the ELEC2 stream. This is a known characteristic of DDM on real-world datasets and represents a limitation that could be addressed in future work by tuning the warning and drift thresholds or applying a warning cooldown period.

Conclusion:

This project implemented and evaluated a sliding window classifier with DDM-based drift detection on the ELEC2 dataset. The adaptive classifier achieved a final prequential accuracy of 80.39%, outperforming the static baseline of 77.28%, confirming that drift-aware adaptation improves performance in non-stationary environments.

DDM successfully detected 144 drift points across the stream, triggering window resets that allowed the classifier to retrain on current data. However, the high warning count (2,936) suggests DDM is sensitive to noise in real-world data, and frequent resets reduced the total instances evaluated, a known tradeoff between adaptation speed and coverage.

Future work could explore threshold tuning, warning cooldown periods, or comparisons against other detectors such as ADWIN or EDDM to further improve robustness.

References

[1] I. Žliobaitė, "Learning under concept drift: An overview," *arXiv preprint arXiv:1010.4784*, 2010.

[2] J. Gama, P. Medas, G. Castillo, and P. Rodrigues, "Learning with drift detection," in *Advances in Artificial Intelligence – SBIA 2004*, A. L. C. Bazzan and S. Labidi, Eds. Berlin, Heidelberg: Springer, 2004, pp. 286–295.